

# PRAVEENA SATTI

AI/ML RESEARCHER | RELIABLE RAG | INFORMATION RETRIEVAL | EVIDENCE VERIFICATION

Hyderabad, India | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

## RESEARCH IDENTITY

**Reliable retrieval-augmented language systems.** I study how changes in retrieved evidence cause answer reversals, and how evidence-aware verification and abstention can prevent harmful transitions. My central methods are information retrieval, transition-based evaluation, controlled failure analysis, and reproducible systems research.

## FLAGSHIP RESEARCH PROGRAM

### From Measurement to Mitigation of Evidence-Induced Answer Reversals in RAG

**Question:** When retrieved evidence is expanded along a fixed ranking, which answer changes are harmful, why do they persist or recover, and can a change-triggered evidence-stability gate reduce correct-to-incorrect transitions without suppressing incorrect-to-correct repairs?

**Status:** Unit tests and the no-cost mock smoke workflow pass. EAR, BCR, RTB, persistence, recovery, and first-reversal metrics are implemented. The primary real-model experiment and mitigation study are pending; mock outputs are not reported as research findings.

[Repository](#) | [27-paper literature map](#) | [Scope and zero-budget plan](#)

## RESEARCH AND OPEN-SOURCE EVIDENCE

### Evidence-Induced Answer Reversals - [code](#)

*Flagship research implementation* | 2026

- Built fixed-prefix trajectory generation, strict input validation, exact-match/token-F1 scoring, bootstrap analysis, and machine-readable result contracts.
- Mapped 27 primary papers and identified retrieval-size robustness, transition metrics under spurious features, sufficient-context selection, and adaptive stopping as required comparisons.

### Castorini Foundations of Retrieval

*Anserini and Pyserini* | 2026

- Reproduced classical, sparse, and dense retrieval runs and contributed evidence logs to established open-source IR toolkits.
- Both contributions passed maintainer review and were merged upstream: [Anserini PR #3391](#) and [Pyserini PR #2650](#).

### Emotion Classification Using BERT: A Comprehensive Study - [record](#)

*Journal of Propulsion Technology*, 44(4) | 2023

- Developed a Transformer-based multi-label emotion-classification workflow on GoEmotions, including preprocessing, BERT fine-tuning, evaluation, and label-level analysis.

## RESEARCH-RELEVANT EXPERIENCE

### AI/ML Engineer Intern - Infor, Hyderabad

2025 - Present

- Built enterprise retrieval-augmented generation pipelines using LangChain, vector indexing, SQL preprocessing, REST deployment layers, and evaluation-oriented data flows.
- Developed NLP automation and reusable AWS data/ML workflows using S3, Lambda, Docker, CI/CD, preprocessing, and monitoring.

### Software Development Intern - Swecha

*Internship*

- Improved backend reliability for an open-access learning platform by auditing API/database retrieval paths, resolving data bottlenecks, and building reusable backend modules.

## PRIMARY STUDY DESIGN

Open QA datasets; one fixed BM25 ranking and one open dense ranking; at least two model families; deterministic prompts; exact-match and token-F1 scoring; bootstrap confidence intervals; and controlled relevance, conflict, order, and support analysis. The gate will report EAR, BCR, RTB, accuracy, abstention coverage, verifier calls, latency, and token overhead. The core path uses open data and open models on available local or free notebook compute; a paid API is not required.

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## CORE SYSTEMS, REPRODUCIBLE ENGINEERING, AND CREDENTIALS

### SELECTED SYSTEMS SUPPORTING THE RESEARCH PROGRAM

|                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hybrid Wiki Search Engine - 10 tests passed</b><br><br>BM25, TF-IDF, reciprocal-rank fusion, and evaluation utilities in a compact retrieval system. <a href="#">Repository</a>                               | <b>PraxisMesh - 9 tests passed</b><br><br>Typed task graphs, deterministic policy gates, human approval, independent postcondition checks, and tamper-evident audit trails. <a href="#">Repository</a>                                                   |
| <b>Repo-Aware Coding Assistant - 6 tests passed</b><br><br>AST chunking, call-graph expansion, lexical retrieval, and traceable file/line grounding for repository-aware suggestions. <a href="#">Repository</a> | <b>Clinical Trial Matching - 8 tests and build passed</b><br><br>Semantic retrieval plus structured eligibility filters. Tests avoid the optional model download; default embedding bootstrap is a separate integration step. <a href="#">Repository</a> |

### PORTFOLIO AUDIT AND ENGINEERING FOUNDATION

**13 executable repositories passed their available test or build checks on 25 August 2026.** Supporting systems cover infrastructure validation, ETL observability, distributed ML execution, graph anomaly detection, language analysis, physiological-signal fusion, synchronized-signal analysis, and authenticated application deployment. These demonstrate testing, observability, controlled execution, and documented boundaries; they are not presented as separate research agendas.

### EDUCATION

#### Bachelor of Technology (Honours) - Computer Science and Engineering

*Artificial Intelligence and Machine Learning | KL University | 2022 - 2026*

- Coursework and independent work across machine learning, deep learning, NLP, information retrieval, databases, algorithms, cloud/data systems, and software engineering.

### RESEARCH METHODS AND TECHNICAL SKILLS

|                    |                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research</b>    | Reliable RAG, information retrieval, answer-trajectory analysis, retrieval robustness, evidence attribution, faithfulness, abstention, failure analysis, and reproducible evaluation.      |
| <b>Methods</b>     | BM25, sparse/dense retrieval, SPLADE, Lucene, reciprocal-rank fusion, BERT/Transformers, PyTorch, Scikit-learn, semantic search, bootstrap confidence intervals, and controlled ablations. |
| <b>Engineering</b> | Python, SQL, Pandas, NumPy, PySpark, FastAPI, Flask, Docker, AWS, Terraform, Neo4j, NetworkX, CI/CD, testing, observability, and Linux.                                                    |

### EARLIER WORK AND EVIDENCE BOUNDARIES

- EMCAD checkpoint screening:** earlier medical-segmentation reproducibility scripts and recorded evaluation evidence; the public branch does not include an automated test suite. [Branch](#)
- BioTwin-PE:** earlier conceptual multimodal monitoring preprint; the public repository is documentation-focused and does not currently contain an executable implementation. [Repository](#)

### CREDENTIALS, AFFILIATIONS, AND RESEARCH PRACTICE

**Certifications:** AWS Certified Cloud Practitioner | Google Associate Cloud Engineer | GitHub Foundations | Neo4j Certified Professional | Automation Anywhere Advanced RPA Professional

**Affiliations:** IEEE Member | IEEE Women in Engineering | IEEE SIGHT

**Practice:** freeze the question and protocol; record dataset/split, retrieval ranking, model and prompt, source revision, environment, raw outputs, and limitations; distinguish passing software checks from scientific findings; report harmful and beneficial transitions together.